

Noah Thompson

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EDUCATION

University of Notre Dame | Notre Dame, IN May 2027
Bachelor of Science | Major: Computer Science | Minor: Sustainability GPA: 3.52

Notre Dame Alcoa Study Abroad Program | Alcoa, Spain May – June 2024

Engineering Projects

Introduction to Artificial Intelligence | University of Notre Dame August – December 2025

- Implemented machine learning methods including decision trees, Markov models, Bayesian reasoning, and logistic regression to perform quantitative evaluation and probabilistic analysis
- Performed supervised and unsupervised learning techniques such as perceptrons, SVMs, KNNs, for predictive modeling, as well as K-Means clustering and PCA for pattern discovery and dimensionality reduction to increase model performance
- Built and trained deep learning models from scratch, including feedforward networks, convolutional neural networks for image-based tasks, recurrent neural networks for sequential data, and transformer-based models in PyTorch to leverage attention

Computer Architecture | University of Notre Dame August – December 2025

- Evaluated processor organization, pipelining, cache and memory hierarchies, and RISC-V instruction set architecture, performing low-level performance analysis and processor design optimizations

Operating Systems | University of Notre Dame August – December 2025

- Developed a heap management implementation with custom malloc using sbrk() and a freelist, improving memory allocation efficiency for user level programs
- Engineered a multi-core process queue shell supporting FIFO and round robin scheduling, demonstrating practical CPU scheduling strategies and process management
- Constructed a concurrent publish–subscribe messaging system using POSIX threads, enabling topic-based subscriptions, message publishing, and blocking retrieval to support a real-time chat application

Systems Programming | University of Notre Dame January – May 2025

- Designed a brute-force password cracking program in Python, implementing process control and inter-process communication to parallelize hash comparisons while analyzing performance tradeoffs in CPU-bound workloads

Data Structures | University of Notre Dame January – May 2025

- Implemented fundamental data structures (linked lists, stacks, queues, trees, and hash tables) in C and Python following Zybook exercises, testing and debugging each different type of data structure while also analyzing time and space complexities

EXPERIENCE

Taylor High School | Kokomo, IN January 2022 – August 2025

Youth Sports Referee

- Officiated 20+ youth basketball games per season, demonstrating strong decision-making skills in high-pressure situations to make accurate calls and ensure fair play
- Fostered a safe and respectful environment for players, emphasizing sportsmanship, teamwork, and positive engagement

RelaxationPro | Kokomo, IN January 2023 – June 2023

CEO

- Founded an e-commerce wellness brand utilizing a scalable dropshipping business model to deliver accessible self-care products
- Instituted the Shopify-based RelaxationPro website, integrating AliExpress dropshipping to automate inventory management, order fulfillment, and secure online transactions

LEADERSHIP AND ACTIVITIES

Society of Latino Engineers and Scientists (SOLES) | Notre Dame, IN August 2024 – Present

- Network with Notre Dame alumni and STEM professionals to expand connections and explore career development opportunities in engineering and science

Transformational Leaders Program | Notre Dame, IN August 2023 – Present

- Participated in a cohort-based mentorship program focused on academic, professional, and leadership development for historically marginalized students, fostering a sense of community and personal growth.
- Engaged in personalized mentoring and community-building activities aimed at developing leadership skills, academic success strategies, and holistic well-being.

TECHNICAL SKILLS

Technical: Python, C, JavaScript, Matlab, HTML, Data Analysis, Data Visualization, Pandas, Matplotlib, Git, Github, Matlab App Designer